Scientific Justification *Be sure to include overall significance to astronomy. For standard proposals limit text to one page with figures, captions and references on no more than two additional pages.*

Young star clusters form embedded within dense molecular gas and dust and subsequently disperse their natal material through stellar feedback processes such as ionizing radiation, stellar winds, and supernova explosions. Constraining the timescale over which this clearing occurs is essential for understanding cluster evolution and the efficiency of feedback: a very short clearing timescale ($\lesssim$ a few Myr) would imply that early feedback processes dominate, while supernovae play a limited role in the initial dispersal of material. In this study, we measure this timescale by combining new Paschen-β observations with archival Hydrogen-α data and cluster ages in NGC 628, directly probing how extinction evolves over the first $\sim$5 Myr of cluster evolution. By targeting clusters in both the central and spiral arm environments, we also assess how this process depends on galactic environment.

A practical method to probe this clearing process is to compare hydrogen lines at different wavelengths. The Hα line at $0.6563\mu m$ is strongly affected by dust attenuation, while the longer-wavelength Paβ line at $1.282\mu m$ is significantly less obscured. The ratio

$$\frac{F_{Pa\beta}}{F_{H\alpha}}$$

therefore serves as a direct tracer of nebular extinction, with more embedded regions expected to exhibit elevated Paβ/Hα ratios. By measuring this ratio for clusters of known age, we can map the decline in extinction as clusters evolve and disperse their natal material.

Recent work has begun to probe this early phase of cluster evolution. Deshmukh et al. (Figure 1) find a rapid $\sim$85% decrease in extinction between $\sim$1 and $\sim$5 Myr in NGC 5236, suggesting that gas clearing occurs very early in cluster lifetimes. However, most previous studies of extragalactic cluster populations (e.g., Bastian et al.; Chandar et al.) largely exclude such young, embedded sources. This program expands on recent work by targeting similarly young clusters in NGC 628, testing whether rapid early clearing is a general feature of cluster evolution across different galactic environments.

NGC 628 is an ideal target for this study. It is a nearby ($\sim$ 10 Mpc), nearly face-on grand-design spiral galaxy with extensive archival data and a rich population of star-forming regions. Its orientation minimizes projection effects and enables spatially resolved studies of individual clusters across distinct galactic environments. As shown in Figure 2, archival HST imaging from surveys such as LEGUS (e.g., Adamo et al.; Shabani et al.) and PHANGS (e.g., Thilker et al.; Buckner et al.; Querejeta et al.; Whitmore et al.) covers 2 fields: the galactic core and the eastern spiral arm, providing high-quality Hα data in each region.

The critical step in converting extinction measurements into a clearing timescale is the use of independent cluster age estimates. Public LEGUS catalogs provide ages and masses for star clusters in NGC 628 derived from broadband HST photometry. These catalogs enable a direct comparison between extinction-sensitive line ratios and cluster age, allowing us to interpret observed trends as part of cluster evolution.

We propose to obtain Paβ imaging of selected young clusters in NGC 628 and combine these data with archival Hα measurements and cluster age estimates. The primary outcome will be a relation between $\frac{F_{Pa\beta}}{F_{H\alpha}}$ and cluster age, from which we will infer the characteristic timescale over which clusters clear their natal gas and dust.

Because the selected targets span both the central region and spiral arms (Figure 2), this proposal will also enable a comparison of clearing behavior across different galactic environments. Variations in star formation conditions between these regions may lead to systematic differences in clearing timescales, providing insight into how environment influences the efficiency of stellar feedback.

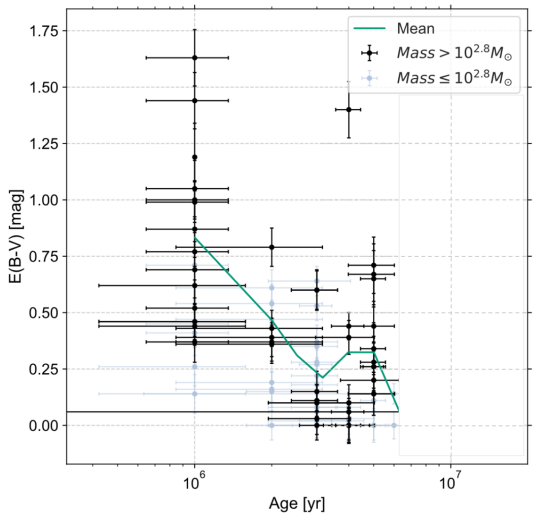


Figure 1: Adapted from Deshmukh et al., showing the evolution of extinction in young star clusters as a function of age for the galaxy NGC 5236. The region shown corresponds to the age range targeted in this work (< 5 Myr). The figure demonstrates a clear decrease in extinction between ∼1 Myr embedded clusters and ∼5 Myr systems that have largely cleared their natal gas. This rapid decline motivates our observations: by measuring Paβ/Hα ratios for clusters in this age range, we aim to directly test whether this early gas-clearing trend holds across different environments within NGC 628.

References

Deshmukh, S., et al. The Astrophysical Journal Letters 974, 24 (2024)

Bastian, N., et al., MNRAS, 419, 2606 (2012)

Chandar, R., et al., ApJ, 787, 17 (2014)

Adamo, A., et al. The Astrophysical Journal 841, 131 (2017)

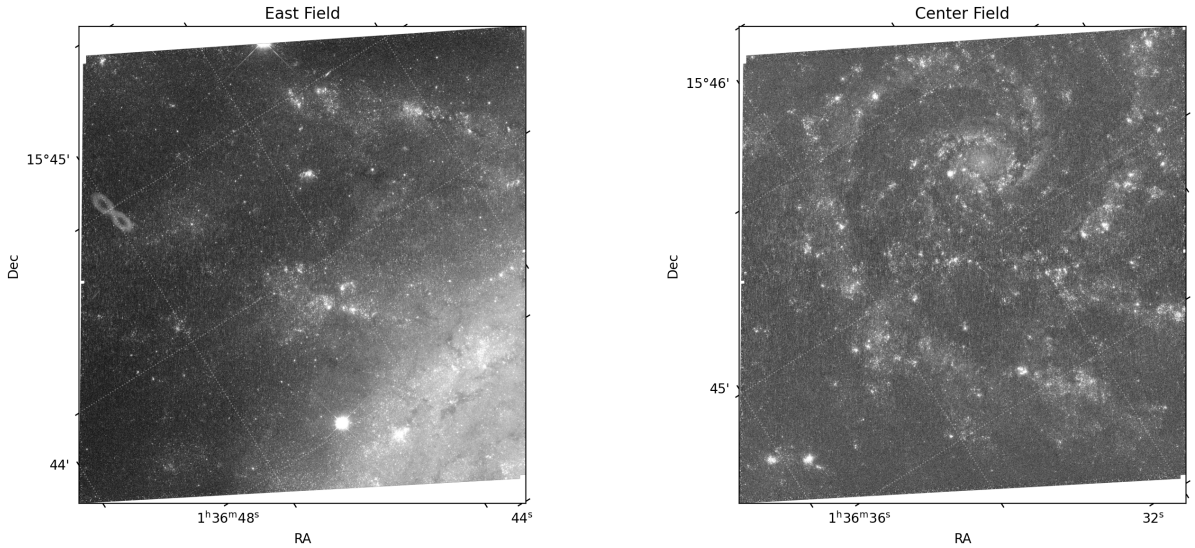
Shabani, F., et al. Monthly Notices of the Royal Astronomical Society 478, 3590 (2018)

Thilker, D. A., et al. Monthly Notices of the Royal Astronomical Society 509, 3 (2022)

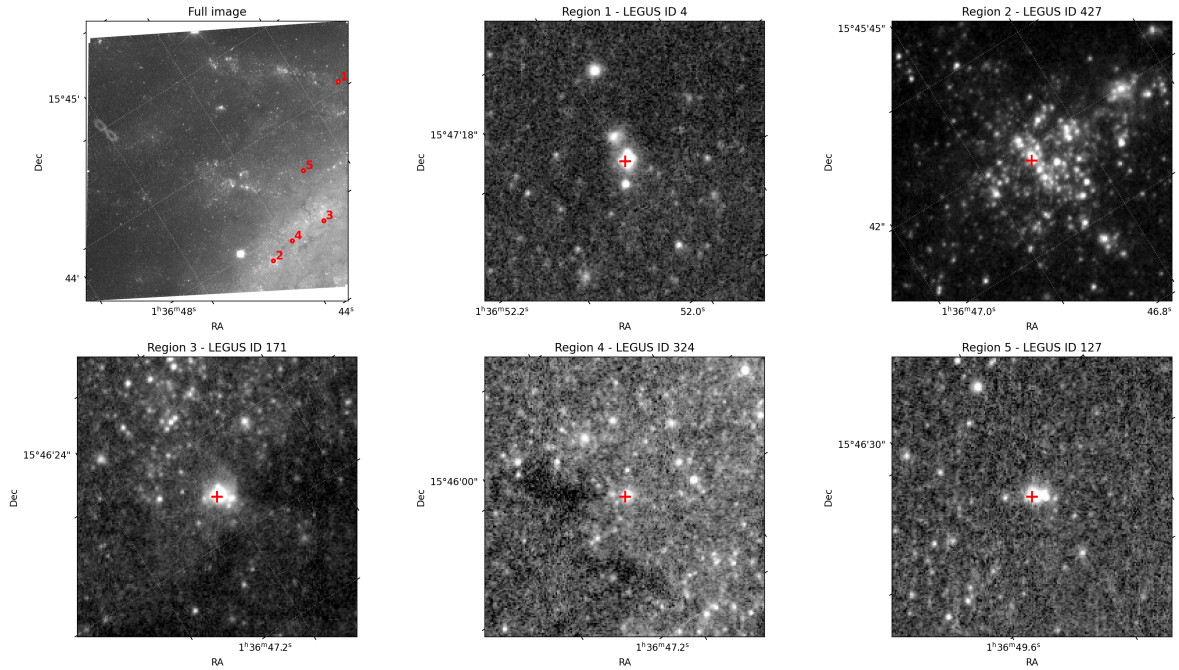
Buckner, Anne S. M., et al. Monthly Notices of the Royal Astronomical Society 545, 2025 (2026)

Querejeta, M., et al. Astronomy & Astrophysics 656, A133 (2021)

Whitmore, B. C., et al. Monthly Notices of the Royal Astronomical Society 520, 63 (2023)



(a) HST imaging of the two NGC 628 fields in the F275W band, showcasing the different galactic environments. The central field probes the galactic inner disk, while the eastern field samples a spiral arm environment, enabling a comparison across distinct galactic conditions.



(b) Zoomed view of the eastern field highlighting the five eastern field star clusters targeted in this proposal. Each cluster is shown in an individual cutout to illustrate the local environments of the targets. These clusters span the key age range (< 5 Myr) and will be used to measure Paβ/Hα ratios, allowing us to probe dust clearing timescales within this galactic environment.

Figure 2: Archival fields and a few targets proposed in this study. The top panels show the two HST fields in NGC 628 (central and eastern), which have the extensive archival Hα data required for comparison with new Paβ observations. The bottom panels highlight the five young star clusters selected in the eastern field, each with a zoomed cutout to showcase their local environments.

Experimental Design *Describe your overall observational program. How will these observations contribute toward the accomplishment of the goals outlined in the science justification?*

Proposed Targets

The proposed targets consist of ten young star clusters in NGC 628, selected from the LEGUS catalogs. These include five clusters in the central field (Table 1) and five in the eastern (spiral arm) field (Table 2). The targets were chosen to be among the highest-mass (and therefore brightest) young systems, while also providing a representative sampling of different galactic environments and a range of young ages, enabling a comparison of cluster evolution and gas clearing across the galaxy.

The LEGUS survey provides cluster ages and masses derived from multi-band HST photometry, while PHANGS-HST provides archival Hα imaging for these regions. Together, these datasets enable a direct comparison between new Paβ observations and existing optical measurements.

Candidate Selection Criteria

NGC 628 was selected as the target galaxy because it is a nearby ($\sim$10 Mpc), nearly face-on spiral galaxy, allowing individual star-forming regions across different parts of the galaxy to be spatially resolved. In addition, it has extensive archival coverage, including HST imaging and cluster catalogs spanning both the central region and spiral arms, making it particularly well suited for combining new near-infrared observations with existing datasets.

Within NGC 628, the final sample of ten clusters was selected to balance environmental coverage, evolutionary stage, and observational feasibility. Five clusters are drawn from the central field and five from the eastern spiral arm, enabling a direct comparison of feedback and gas-clearing behavior across distinct galactic environments. All selected clusters are young ($\lesssim$ 5 Myr), ensuring they are in the embedded or partially embedded phase where dust clearing is actively occurring; this is also the age range over which recent studies (e.g., Deshmukh et al.) have identified rapid extinction evolution, making it critical to probe this regime. Finally, the clusters are preferentially high-mass systems, increasing the likelihood of strong nebular emission and detectable Paβ flux while minimizing the required observing time.

LEGUS ID	RA (deg)	Dec (deg)	Age (Myr)	$\log(M/M_\odot)$
591	24.1637	15.7951	2.4	3.93
928	24.1523	15.7893	3.2	3.88
655	24.1603	15.7936	4.3	3.79
899	24.1602	15.7899	4.3	4.08
388	24.1794	15.8004	5.1	3.74

Table 1: Selected young star clusters in the central region of NGC 628.

LEGUS ID	RA (deg)	Dec (deg)	Age (Myr)	$\log(M/M_\odot)$
4	24.2179	15.7892	2.0	3.74
427	24.1965	15.7629	2.6	3.93
171	24.1979	15.7741	2.9	3.93
324	24.1978	15.7676	3.2	3.93
127	24.2076	15.7757	4.7	3.74

Table 2: Selected young star clusters in the eastern (spiral arm) region of NGC 628.

Signal To Noise Ratio Requirement

The primary observable in this program is the ratio $\frac{F_{Pa\beta}}{F_{H\alpha}}$, which serves as a tracer of nebular extinction and its evolution with cluster age. To distinguish between embedded and cleared clusters, the Paβ flux must be measured with sufficient precision to constrain this ratio.

We aim to detect a trend in extinction similar to that observed by Deshmukh et al. (Figure 1), which shows a strong decline in extinction over the first $\sim$5 Myr. To robustly identify the presence (or absence) of such a trend across our sample, we require fractional uncertainties of $\sim$11% in the Paβ flux measurements. This corresponds to a signal-to-noise ratio of $\sim$9 per target. Achieving a comparable signal-to-noise ratio in the adjacent continuum is also necessary to ensure reliable continuum subtraction and accurate line flux recovery.

Instrument Choice and Exposure Time Calculation

Observations will be conducted using Gemini North/NIRI in imaging mode with the Paβ narrowband filter (central wavelength $1.282\mu m$) and a nearby J broadband filter ($1.25\mu m$) for continuum subtraction. The f/6 camera will be used, providing a pixel scale of $0.117''$/pixel and a field of view of $120'' \times 120''$. This configuration is optimized for observations at wavelengths $\lesssim 2.5\mu m$ and is well matched to the spatial scales of the selected clusters.

Using the Gemini Integration Time Calculator, we find that an exposure time of 60 seconds per target in the Paβ narrowband filter is sufficient to achieve the required signal-to-noise ratio of $\sim$9. For the J-band observations, an exposure time of 5 seconds per target yields a comparable signal-to-noise ratio.

The ITC also provides the relevant overheads, which are similar for both filters. These include a setup time of 360 seconds, a readout time of 1 second, and a DHS write time of 5.1 seconds per exposure. Accounting for all exposures and overheads across the ten targets results in a total telescope time of 2 hours, 12 minutes, and 50 seconds.

ITC Examples Shown below is a Gemini/NIRI Integration Time Calculation for one of the sources.

Gemini Integration Time Calculator
NIRI - 2026B.1.1.1

Click here for help with the results page.
software aperture diameter = 0.90 arcsec
fraction of source flux in aperture = 0.62
enclosed pixels = 47.59

derived image size (FWHM) for a point source = 0.77 arcsec.

Contributions to total noise (e-) in aperture (per exposure):
Source noise = 76.15
Background noise = 406.22
Dark current noise = 26.72
Readout noise = 82.79

Total noise per exposure = 422.35
Total signal per exposure = 5798.53

Derived number of exposures = 1, of which 1 is on source.
Taking 1 exposure, the effective S/N for the whole observation is 9.79 (including sky subtraction)

Required total integration time is 60.00 secs, of which 60.00 secs is on source.

The peak pixel signal + background is 3680 e- (299 ADU). This is 2% of the full well depth of 200000 e-.

Observation Overheads	
Setup	360.0 s
Exposure	1 x 60.0 s
Readout	1 x 1.0 s
DHS Write	1 x 5.1 s
Program time	**7 mins 6 secs**

Input Parameters:
Instrument: NIRI

Source spatial profile, brightness, and spectral distribution:
The z = 0.00000 point source is an emission line at a wavelength of 1.282 µm with a width of 100.0 km/s.
It's total flux is 5.0E-19 W/m² on a flat continuum of flux density 1.0E-16 W/m²/µm.

Instrument configuration:
Optical Components:
- Filter: NBF_PABETA
- Fixed Optics
- Camera: f6
- Detector - 1024x1024-pixel ALADDIN InSb array
- Read Mode: 1-2.5um: Faint Object Narrow-band Imaging/Spectroscopy
- Detector Bias: shallow well

Pixel Size: 0.116

Telescope configuration:
- silver mirror coating.
- up looking port.
- wavefront sensor: pwfs

Observing Conditions:
- Airmass: 1.50
- Image Quality: 70% (≤ 0.60" at zenith, ≤ 0.77" on-source)
- Cloud cover: 70%
- Water Vapor: 100%
- Sky Background: 100%

Likelihood of execution: 49%

Calculation and analysis methods:
- Mode: imaging
- Calculation of integration time from a S/N ratio of 9.00 for exposures of 60.00 secs, and 100.00% of them on source.
- Analysis performed for aperture that gives 'optimum' S/N and a sky aperture that is 1.00 times the target aperture.